BiNeuroRAM: Energy-Efficient ReRAM-Based PIM for Accurate Bipolar Spiking Neural Network Acceleration

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***Abstract*—ReRAM is a promising non-volatile memory for neuromorphic accelerators, yet it faces challenges such as high sensing power and accuracy degradation. This work proposes BiNeuroRAM, a novel spiking neural network (SNN) accelerator leveraging ReRAM-based processing-in-memory (PIM), with three key contributions: (1) It is the first to support higher-accuracy spike-tracing bipolar-integrate-and-fire (ST-BIF) neurons, achieving 80.9% accuracy on ImageNet, 8.4% higher than the previous state-of-the-art; (2) It introduces a low-power voltage sense amplifier (LPVSA) that reduces ReRAM read power by 14.7∼58.2×, enhancing energy efficiency; (3) It employs an asynchronous micro-architecture that fully exploits the event-driven nature of SNNs. Experimental results show that BiNeuroRAM improves throughput density and energy efficiency by 2.08× and 2.09× on ImageNet with ResNet-18, compared to traditional integrate-and-fire (IF) neuron-based SNN accelerators.**

***Index Terms*—Neuromorphic hardware, spiking neural network, processing-in-memory, asynchronous circuit, bipolar neuron**

I. Introduction

Recent advances in artificial intelligence (AI) have driven the demand for high-performance and low-power algorithms, making spiking neural networks (SNNs) a promising solution for energy-efficient AI systems [1]. SNN emulates the communications of neurons in the human brain to propagate with discrete asynchronous events (*i.e.*, spikes), enabling significant energy savings over traditional artificial neural networks (ANN). This potential attracts attention from both industry [2], [3] and academia [4], [5], accelerating the development of SNN-based neuromorphic hardware.

Prior studies primarily focus on the hardware [2]–[10] and algorithm [11]–[14] designs, which have achieved prominent advances in terms of performance (throughput, efficiency, *etc.*) and accuracy. [6]–[10] leverages processing-in-memory (PIM) architecture to break the "memory wall" and mitigate the communication overhead, leading to boosted performance over the ASIC accelerators [2]–[4]. Among the PIM-based neuromorphic accelerators, ReRAM-based designs [8], [9] present higher density, while SRAM-based designs [6], [7] achieve lower energy for the ADC-free charge-based computing scheme. The SNN algorithms, such as IF [11] and leaky-IF (LIF) [12] neurons, are continuously optimized to reduce energy and latency [11]–[14].

Despite the success of prior works, our investigations reveal that challenges remain in neuromorphic architecture design. First, existing PIM-based designs [6]–[10] predominantly adopt IF/LIF neurons, which suffer from over-firing issues [14]. The membrane potential of an IF/LIF neuron is hard-reset after the neuron firing spikes, leading to information loss and conversion mismatches (detailed in Section II-B), ultimately resulting in significant accuracy degradation. Second, while ReRAM-PIM [9] improves parallelism and throughput using small-sized arrays, they incur substantial area and power overhead by analog-digital-converters (ADCs). Although current sense amplifiers (CSA) are employed in [8] to eliminate ADC and reduce access time in large arrays, their benefits are marginal in small arrays [15], and continuous current sensing leads to increased power consumption. Lastly, most existing PIM-based SNN accelerators adopt synchronous architectures, where continuous clocking for membrane potential updates further contributes to excessive power consumption.

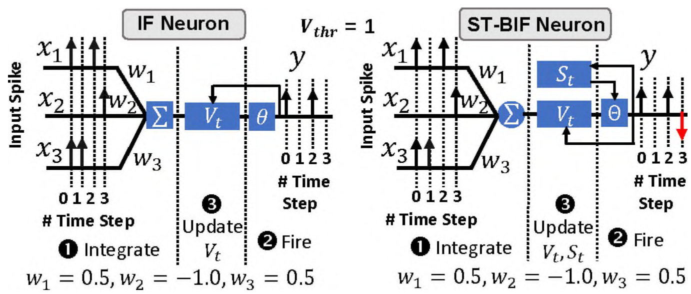


Fig. 1: The neural dynamics of IF and ST-BIF neurons. ST-BIF neuron fires bipolar spikes to eliminate overfiring imperfection [14] and prevent accuracy degradation.

As a result, a more energy-efficient SNN accelerator with higher inference accuracy is highly desired. To this end, three key challenges must be addressed: 1) The limited computational expressiveness of IF/LIF neurons constrains scalability and hinders accuracy improvements; 2) Although small-sized ReRAM arrays enhance throughput, both ADC-based analog computing [9] and CSA-based digital computing [8] impose significant energy overheads; 3) As the system scales, reliance on power-hungry global clocks becomes a major bottleneck, conflicting with the inherently asynchronous behavior of SNNs.

This work proposes BiNeuroRAM, a novel asynchronous ReRAM-based PIM architecture that supports SNNs with emerging spike-tracing bipolar integrate-and-fire (ST-BIF) neurons [14], targeting high accuracy and energy efficiency. Unlike conventional hard-reset IF/LIF neurons, ST-BIF neurons require soft-reset, which is fundamentally incompatible with the hard-reset analog charge-based accumulation in [6]–[8]. BiNeuroRAM retains the analog charge-based accumulation from [16] but replaces the power-hungry ADC with asynchronous self-trigger converters (STC). Furthermore, we design a low-power voltage sense amplifier (LPVSA) dedicated to the small-sized ReRAM array sensing offering a compact and efficient design. Compared to SOTA SNN-based neuromorphic designs [2], [7]–[9], BiNeuroRAM achieves superior performance, the highest accuracy, and the lowest latency. Our contributions are:

- BiNeuroRAM is the first neuromorphic hardware that realizes ST-BIF neurons with delicate circuit designs. Negative spikes are supported to neutralize the over-firing energy without breaking the ANN-SNN conversion equivalence, thereby improving accuracy.
- BiNeuroRAM features a novel asynchronous micro-architecture

with an LPVSA tailored for energy-efficient PIM operations on small-size ReRAM arrays, enabling high parallelism and improved throughput. The ADCs are replaced by asynchronous self-trigger converters (STC), reducing area and power consumption.
- To fully exploit the event-driven nature of SNNs, BiNeuroRAM introduces sparsity-aware optimization on both input spikes and weight bits, reducing the power and deriving better efficiency.
- Experiments demonstrate that BiNeuroRAM outperforms state-of-the-art neuromorphic accelerators. BiNeuroRAM achieves 8.36% higher accuracy on ImageNet with ResNet-50, along with 2.08× higher throughput density and 2.09× better energy efficiency.

II. Background and Related Work

A. Spiking Neural Network

1) Introduction to SNN: Spiking Neural Networks (SNNs) are biologically-inspired neural networks that use time-dependent spikes for communication, offering energy efficiency and temporal dynamics, suitable for neuromorphic computing and real-time, event-driven tasks.

2) SNN Neuron Models: IF neuron is one of the simplest models and widely applied in neuromorphic accelerators [2]–[5]. However, "over-firing" (firing redundant spikes) [13] is introduced by the IF neuron, resulting in accumulated output errors and accuracy degradation. Besides, overfiring can incur additional power consumption and decrease the overall energy efficiency. LIF eliminates the overfiring by membrane potential leaking, but breaks the conversion equivalence and suffers from severe accuracy loss. To overcome the challenge, ST-BIF neuron is proposed [13], [14] to neutralize the over-fired positive spikes with negative ones thus strictly maintaining the conversion equivalence, as elaborated in Fig. 1. The dynamics of ST-BIF neuron are:

$$\begin{aligned} V_t &= V_{t-1} + V_t^{\mathrm{in}} - V_{\mathrm{thr}} \cdot \Theta(V_{t-1} + V_t^{\mathrm{in}}, V_{\mathrm{thr}}, S_{t-1}) \\ S_t &= S_{t-1} + \Theta(V_{t-1} + V_t^{\mathrm{in}}, V_{\mathrm{thr}}, S_{t-1}) \\ \Theta(V, V_{\mathrm{thr}}, S) &= \begin{cases} 1; & V \geq V_{\mathrm{thr}} \ \& \ S < S_{\max} \\ 0; & \text{other} \\ -1; & V < 0 \ \& \ S > S_{\min} \end{cases} \end{aligned} \tag{1}$$

where V_t is the membrane potential of the neuron at time-step t; V_{thr} is the threshold voltage for a neuron to fire a positive spike; S_t is the value of spike tracer at time-step t; V_t^{in} is the input voltage at time-step t; Θ is the firing behavior function of ST-BIF neuron.

Fig. 1 presents the example neural dynamics of both vanilla IF and ST-BIF neurons. The ST-BIF neuron differs from the vanilla IF neuron from two perspectives: (1) ST-BIF neuron fires bipolar (positive/negative) spikes; (2) ST-BIF neuron ensures the accumulated charge of output spikes always equals the output of the Q-ReLU function until the firing stops, in virtue of the spike tracer S_t.

3) SNN Applications: SNNs have been applied to various datasets, showcasing their strengths in energy-efficient and real-time processing for static vision task, neuromorphic vesion task, and natural language processing (NLP) task. In image recognition [13], [14], SNNs especially with ST-BIF neuron performs well on datasets like MNIST, CIFAR-10, and ImageNet, achieving competitive accuracy while being power-efficient. They excel in event-based vision tasks using datasets like CIFAR10-DVS [14] for gesture recognition, leveraging asynchronous event-driven computation. VISTREAM [17] as ST-BIF neuron SNN is also an energy efficient solution in visual streaming perception (VSP) task. While in Natural Language Processing (NLP) tasks, such as text classification and machine translation are available with spike-based models like IF neuron based Spikformer [18] and ST-BIF neuron based SpikeZIP-TF [14]. Their ability to handle temporal and event-driven data makes them ideal for low-power, real-time applications and competitive with ANNs solutions.

B. PIM for SNN Acceleration

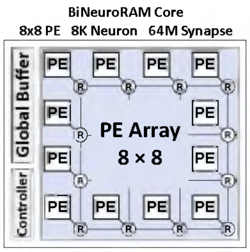
PIM architecture for SNN acceleration has been widely explored and demonstrated to achieve remarkable performance [6]–[10]. Prior PIM [6]–[8] designs leverage analog charge-based neuron circuits by accumulating the membrane potential by charging the capacitor, comparing it with threshold voltage via a converter, and finally hard-reset (discharge) to zero after firing a spike. Such hard-reset-based design does not require ADC, however suffers from a significant accuracy degradation [13], [14]. Neuro-CIM [7] performs in-SRAM multiplication and charge-based accumulation, but is still power hungry due to the clock-driven comparator. Tempo-CIM [8] proposes charge-pump-based neuron circuits and uses time-domain converters to perform SNN operations but sacrifice SNN accuracy. [6] presents a clock-free architecture using a capacitor-based integration circuit but is limited to 4-bit weight accumulation only. NeuRRAM [9] proposes an analog neuron that supports various AI models with forward, recurrent, and backward dataflow, however using an area-power-expensive ADC. ANP-I [10] proposed a digital PIM with asynchronous design to achieve outstanding energy efficiency. Nevertheless, it is only demonstrated with toy applications with a limited number of synapses.

C. Motivation

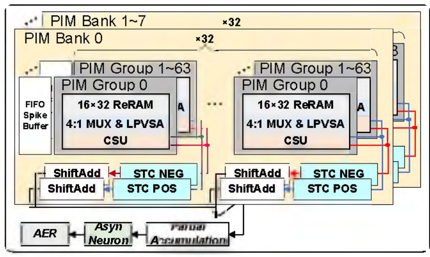
1) Neuron Accuracy: Enhancing the accuracy of SNN algorithms is crucial for realizing their full potential in real-world applications, where energy efficiency, real-time processing, and high accuracy are essential. Addressing challenges such as spike-based computation and over-firing issues, the novel ST-BIF neuron models, when combined with advanced learning techniques, can significantly improve SNN performance, enabling them to achieve or even exceed the accuracy of conventional neural networks.

2) ReRAM-Based PIM: ReRAM is an emerging non-volatile memory that offers advantages like high density, low power, and non-volatility, making it ideal for PIM architectures where computation is performed directly in memory, minimizing data movement and energy consumption. However, optimizing several aspects of ReRAM-based PIM is crucial for improving overall system efficiency, scalability, and performance. 1) Enhancements in read operation, including optimizing sense amplifiers for better power efficiency and accuracy, are necessary to handle small voltage swings and reduce power consumption during memory access. 2) Addressing variability in resistance states, reducing write endurance issues, and improving parallel processing capabilities are critical for maximizing energy efficiency and ensuring robust performance in large-scale SNN neuromorphic systems. These optimizations lead to energy-efficient, scalable, and real-time AI systems for edge computing, robotics, and neuromorphic applications.

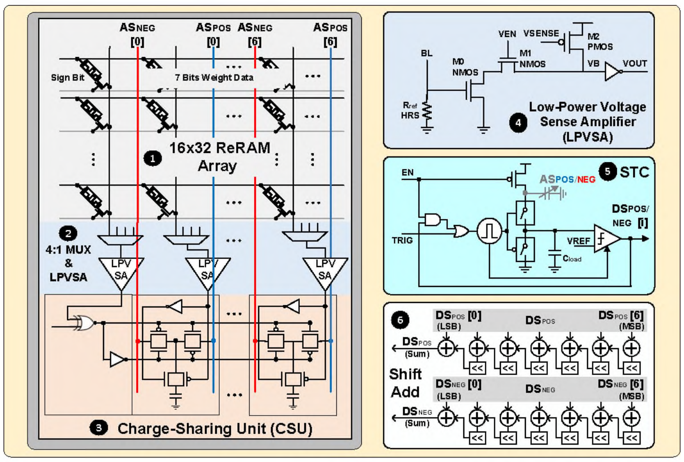
3) Asynchronous Step-by-Step SNN: Asynchronous step-by-step processing in SNN-based hardware lies in improving efficiency, scalability, and real-time performance. In traditional synchronous systems, all neurons update at fixed time intervals, leading to inefficient power usage and unnecessary computations. Asynchronous step-by-step processing allows neurons to spike and update only when triggered by an event, such as reaching a threshold, making it event-driven. This reduces power consumption, as neurons process information only when necessary, and minimizes latency by responding immediately to spikes, without waiting for a clock cycle. Additionally, this approach scales well with larger networks by enabling parallel, distributed computation, making it suitable for real-time applications like robotics, sensory processing, and neuromorphic computing. In summary, asynchronous step-by-step SNN hardware can achieve high



(a) BiNeuroRAM Core



(b) BiNeuroRAM PE



(c)

Fig. 2: (a) Architecture overview of BiNeuroRAM core, where 64 processing elements (PEs) are routed using a mesh-based network-on-chip (NoC); (b) Micro-architecture of BiNeuroRAM PE with 1024×1024 ReRAM array PIM bank, FIFO spike buffer, computing peripherals, and neuron. The column-wise in-memory compute units are connected to STC and output the membrane potential; (c) Detail circuit design of PIM group, LPVSA, STC, and ShiftAdd.

energy efficiency and faster response time, making it ideal for low-power AI applications.

III. BiNeuroRAM Architecture

The architecture overview of BiNeuroRAM is depicted in Fig. 2(a), along with the computation dataflow in Fig. 3. The BiNeuroRAM core consists of a global buffer, a controller, and 64 PEs connected using a mesh-based NoC. The global buffer temporarily stores the inputs and outputs, while the detailed configurations are given in Table I. In summary, one BiNeuroRAM Core supports 8K neurons and up to 64M synaptic weights. Note that BiNeuroRAM takes inference style of step-by-step (*aka.* multiple-time-step) asynchronously, which means the neuron membrane will constantly accumulate every time step across all SNN layers.

A. Hybrid Dataflow of BiNeuroRAM Architecture

In BiNeuroRAM architecture, PIM is leveraged using ReRAM to store weight values, enabling efficient and low-power neural computations. As illustrated in the left figure of Fig. 3, the weights of each kernel are stored in a stationary manner within the ReRAM cells. These weights are only accessed and read when the input streaming activates the corresponding wordline (WL). This weight stationary dataflow approach maximizes weight reuse by keeping the weights fixed in place, significantly reducing the energy-intensive operations associated with repeatedly moving weight data between memory and processing units.

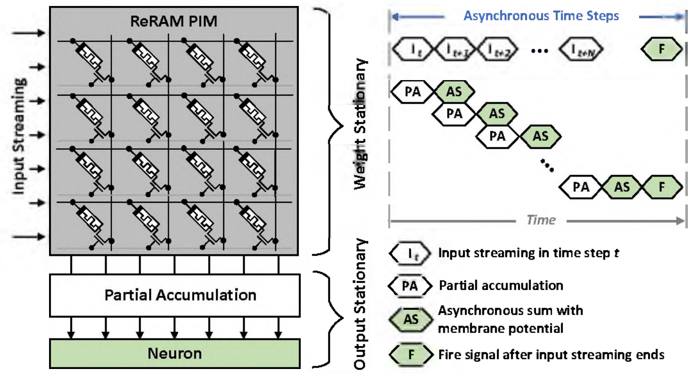


Fig. 3: Dataflow overview of BiNeuroRAM architecture, where it adapts weight stationary dataflow using PIM with output stationary dataflow to constantly accumulate the membrane potential before firing a spike.

Moreover, BiNeuroRAM operates in an inference style that is step-by-step and asynchronous. During this process, the membrane potential in the neuron is accumulated iteratively as the input streaming progresses. The inference continues until the entire set of input streaming is complete, at which point a spike is fired. This firing mechanism aligns naturally with the principles of output stationary dataflow, where the outputs (or membrane potentials) remain stationary in the neuron while the inputs are streamed through the system. The right figure of Fig. 3 illustrates this process in detail; each input stream accesses the ReRAM PIM to read the corresponding weights. These weights are partially accumulated, and the results are accumulated again to update the membrane potential in the neuron. This iterative process continues until all input streams have been processed. Once the computation is complete, a fire signal is generated and sent to the neuron. Upon receiving this signal, the neuron evaluates the accumulated membrane potential against the constraints defined in Eq. (1) and fires a spike if the conditions are met.

B. Microarchitecture of BiNeuroRAM PE

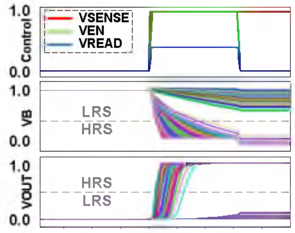
For each BiNeuroRAM PE as illustrated in Fig. 2(b), it is composed of 8 PIM banks, 128 ways accumulators (*i.e.*, 16 ways per bank), and 128 neurons (*i.e.*, 16 neurons per bank). PIM banks are designed to perform weight-stationary SNN inference in a mixed-signal fashion. The accumulator is used to accumulate the partial sums generated by the PIM banks, and then the neurons are leveraged for output firing. An address event representation (AER) circuit converts the output neuron spike events to spike address representation.

1) PIM Bank: In Fig. 2(b), it shows each 8 PIM bank is made up with 64 × 32 PIM groups, while each PIM group is composed by 16 × 32 1-transistor-1-ReRAM (1T1R) bit cells sub-array, leading to a total capacity of 128-kB (1024×1024). All the sub-arrays are sensed via our ❹ low-power voltage sense amplifier (LPVSA). Note that within one PIM bank, PIM groups are physically halved by row driver for sufficient drivability to activate WLs. In addition, the PIM bank consists of 448-ways self-trigger converter (STC) and 16 ways of Shift and Add units (ShiftAdd). Fig. 2(c) shows the detail circuit of ❺ STC and ❻ ShiftAdd.

2) PIM Group: The PIM group plays a major role in handling charge-sharing matrix multiplication (MM) operations. Fig. 2(c) shows the detail circuit of PIM group. PIM group consists of 3 major parts for enabling the charge-sharing process: ❶ 16 × 32 ReRAM array, ❷ 4:1 MUX & LPVSA, and ❸ charge-sharing unit (CSU).

Corner	TT	SS	FF
R_{read}	LRS / HRS		
Latency (ns)	0.8	1.2	0.3
Power (uW)	1.8 / 0.066	1.64 / 0.056	5.09 / 1.03
Energy (fJ)	1.44 / 0.053	1.97 / 0.067	1.53 / 0.31

(a) Evaluation



(b) Monte Carlo Analysis

Fig. 4: LPVSA (a) evaluation over different corners and (b) Monte Carlo analysis for accurate functionality.

❶ 16×32 **ReRAM array** will store 4 different signed 8-bit synaptic weights data in a single row. The first 4 columns store the sign-bit of the 4 weights, while the remaining columns store the 7-bit unsigned integer (*i.e.*, representing $-127 \sim 127$). This design can provide a dense memory capacity and allow sufficient space for our sensing circuit.

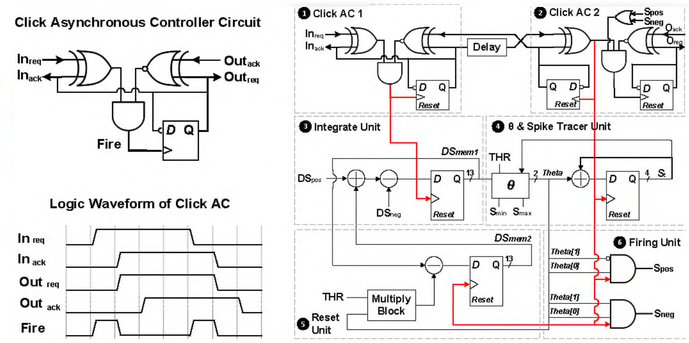
❷ **4:1 MUX** & **LPVSA** will select the correct ReRAM array column based on the AER spike input to sense data. The different currents generated by the varied conductance of the HRS/LRS ReRAM cells will be sensed by our proposed LPVSA connected at the end of the BL. The details of our proposed LPVSA will be described in ❹.

❸ **Charge-sharing unit (CSU)** is integrated into the PIM group to allow charge-sharing operation. It consists of two parts: multiplication unit and 6T1C charge decision unit. The multiplication unit includes an XNOR-gate connecting both signed bits of activation and weight. This helps to determine the polarity of the accumulated results and allows the charge decision unit to activate the corresponding capacitive analogy-summation (AS) wire: blue positive-polarity (AS_{pos}) or red negative-polarity (AS_{neg}).

C. Sensing Circuits

1) Low-Power Voltage Sense Amplifier (LPVSA): As shown in ❹, LPVSA is proposed for a small ReRAM array to replace CSA in heavy load BL. The proposed LPVSA utilizes the discharge voltage on the M2 PMOS and then senses via a low static current path by the M0 NMOS. Firstly, the M2 PMOS is disabled by VSENSE to precharge the sensing circuit to V_{DD}. When an input spike enters, M1 NMOS will pull up by VEN to stop charging and M2 PMOS will pull down by VSENSE signal to start sensing. Meanwhile, the sensing reference resistance R_{ref} is set to HRS as a voltage divider. Since the M0 NMOS are adjusted to a small size, a weak voltage level of ≈400mV (VREAD) can pull up the transistor for grounding when the input is LRS. Finally, the sensing process is completed and the inverter amplifies the precharged voltage, VB for output results, VOUT. If the input is HRS, the gate voltage of M0 is 200mV, which is insufficient to drive the M0 NMOS. Fig. 4a evaluated the performance of our LPVSA when sensing HRS/LRS over different corners. We also completed Monte Carlo analysis to ensure accurate functionality of LPVSA, as shown in Fig. 4b.

2) Self-Trigger Converter (STC): A ❺ STC [19] is integrated to eliminate the usage of the power-hungry clock from ADC when conducting MM operation. From Fig. 2(b), AS_{POS} and AS_{NEG} conduct accumulation and conversion using two different STCs ($STC\ POS$ and $STC\ NEG$). Initially, in the absence of triggering either the start EN or TRIG signal, V_{DD} will precharge the $C_{\text{AS(pos)}}$ and $C_{\text{AS(neg)}}$, while C_{load} will reset to V_{SS}. When there is a spike activity, the pulse generator is triggered and stops the charging-resetting process



(a) Asynchronous Controller

(b) ST-BIF neuron circuit design

Fig. 5: (a) Click-based asynchronous controller integrated into the (b) ST-BIF neuron circuit design with asynchronous clock.

of $C_{\text{AS(pos)}}$, $C_{\text{AS(neg)}}$, and C_{load}. Then, it starts comparison between V_{AS} (V_{load}) and V_{ref}. If V_{load} is greater than V_{ref}, the pulse generator will trigger again for the next comparison. While C_{load} will reset and charge-share with C_{AS} again until V_{ref} is larger than V_{load}, then a stop signal is passed to the next stage. A maximum of 64 conversions will be imposed if all 64 capacitors are activated in a single AS line. The number of conversions at the $STC\ POS$ and $STC\ NEG$ also referred to the digital output of $C_{\text{AS(pos)}}$ and $C_{\text{AS(neg)}}$ activated. Lastly, the total conversion pulses from STC will pass to ShiftAdd to accumulate the total membrane potential.

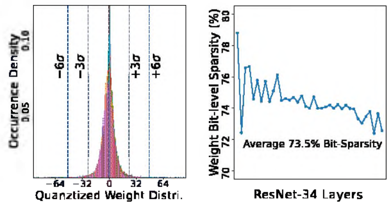
D. ST-BIF Neuron Circuit

1) Asynchronous Controller: Instead of using a synchronous global clock, hand-shake signals in asynchronous logic are more energy efficient and compatible with the multiple-time-step data flow for SNN. Fig. 5a shown the design and logic waveform of Click [20] asynchronous controller (AC). The *req* and *ack* handshake signals are used for data transmission between the sender and receiver. The sender transmits the *req* signal when data are available, while the receiver captures the data and returns an *ack* signal. In the logic waveform, when In_{req} toggles, indicating a valid input event, the input channel fills up, and the fire signal is activated. The rising edge of the fire signal then triggers the DFF to invert the Out_{req} and In_{ack} signals, which resets the phase-decoupled Click signal to its initial state.

2) Neuron Circuit: As shown in Fig. 5, the neuron circuit is designed to mimic the ST-BIF bipolar neuron activity based on the neuron model described in Eq. (1). Both ❶ AC1 and ❷ AC2 will adopt a handshake signal to ensure data synchronization. When ❶ AC1 detects In_{req} signal, ❸ integration unit will accumulate the new signed DS_{POS} and DS_{NEG} as the digital membrane potential, DS_{mem1}. Then, DS_{mem1} enters the θ block in ❹ θ & spike tracer unit to check the firing conditions in Eq. (1). A *Theta* value of signed 2b'01 will be generated if $DS_{\text{mem1}} >$ THR and $S_{\text{t}} < S_{\text{max}}$; a *Theta* value of 2b'11 will be generated if $DS_{\text{mem1}} < 0$ and $S_{\text{t}} > S_{\text{min}}$; Otherwise, a *Theta* value of 2b'00 will be generated when no conditions are satisfied. The ❷ AC2 will fire the DFF in ❹ to record the theta value (spike fired) in the 4-bit spike tracer unit as S_{t}. If a non-zero theta value is generated, it will enable ❺ reset unit for soft-reset by adding or deducting DS_{mem1} with the threshold value, THR. The ❷ AC2 fire signal will control the DFF for updated membrane potential, DS_{mem2}, and pass to ❸ for the next integration. In the meantime, the non-zero theta value and ❷ AC2 fire signal conduct AND3 operation at the ❻ firing unit to fire positive, S_{pos} or negative, S_{neg} spike output. Once the

Models	Spike Ratio (%)		
	-1	0	+1
MLP	0.62	90.1	9.28
ResNet12	1.5	94.2	4.3
ResNet18	4.7	84.3	11.0
ResNet20	3.8	86.9	9.3
ResNet34	4.1	86.0	9.9
ResNet50	6.9	80.6	12.5
Average	3.6	87.0	9.4

(a)



(b) (c)

Fig. 6: (a) Ratio of -1/0/+1 spikes in four representative models; (b) Distribution of 8-bit quantized ResNet-34 weights; (c) Layer-wise bit-sparsity of quantized ResNet-34 weights.

neuron operation is completed, ❷ AC2 will pass the output request, O_{req} signal to the next AC receiver.

E. Sparsity Optimization

1) Input Sparsity: Since SNN inference operation is sparse and event-driven, a power gating method is applied to the CSU to keep the CSU in idle status when the input spikes (activations) are absent. According to our profiling results on the classical ResNet model [21] listed in Fig. 6a, the input spikes are sparse for 87.0% on average. Such power gating can lead to an average of 9.2× power reduction.

2) Weight Sparsity: Since the ReRAM read circuit consumes different energy *w.r.t* the resistance of the ReRAM cell, mapping the HRS (lower power) to more frequently occurring data can significantly reduce the average power. We analyze the data distribution of 8-bit quantized ResNet-34 networks and obtain the ratio of bit-0 and 1, as shown in Fig. 6c. According to the chart, the frequency of bit-0 is 2.8× higher than bit-1 on average. Therefore, should be mapped to lower power consumption HRS.

IV. Evaluation

A. Experimental Setup

We conclude all the key elements below for our BiNeuroRAM architecture and build a dedicated in-house cycle-accurate simulator to evaluate the overall performance. The digital circuits are designed in RTL and synthesized using Synopsys Design Compiler with 28nm standard cell library. We use Cadence Innovus to perform place-and-route, Synopsys VCS for gate-level verification, and Synopsys PrimeTime PX for power analysis. The analog modules are designed and verified using Spectre simulator, with the area obtained at the layout level. The analysis of the NoC modules is conducted using ORION [22] with a 32-bit bus width. Note that, the evaluation of ORION is of 32nm technology originally, and we scale the results to 28nm in this work. The circuit model of the ReRAM model is based on our experimentally-calibrated model [23] in Verilog-A. NVSim [24] is leveraged to model the memory architectural performance, including the read/write latency and energy on the data path. The buffer is evaluated using a foundry 28nm memory compiler. Further details of the BiNeuroRAM configurations and specifications are summarized in Table I.

B. Dataset and Baseline

The SNN parameters for different datasets that are used for inference evaluation along with the network models, namely MNIST [25] 512-256-10 MLP (accuracy and latency only), CIFAR-10 ResNet-12 (W1), CIFAR-10 ResNet-20 (W2) [26], ImageNet ResNet-18 (W3), ImageNet ResNet-34 (W4), and ImageNet ResNet-50 (W5) [27]. All the SNN models are pre-trained via the ANN-to-SNN conversion training method introduced in [14]. For SNN inference, the input spikes are analog-coded [14] with inference style of step-by-step

TABLE I: BiNeuroRAM configurations and specifications.

Component	Configuration	Area (mm^2)	Power (mW)
PIM Bank Spec.			
ReRAM	1024×1024 1-bit/cell LRS: 15kΩ HRS: 265kΩ	0.0088	0.0865
Compute Logic	64×128	0.0126	0.0526
STC	448	0.0085	0.0339
PIM PE Spec.			
Bank	8	0.238	1.384
Neuron	128	0.0433	0.0285
Accumulator	128	0.0035	0.0021
Router	Weight: 32-bits	0.0081	1.489
BiNeuroRAM Core Spec.			
PE	64	18.8	185.8
NoC	Topology: Mesh Weight: 32-bits	0.074	0.0013
Controller	-	1.125	31.44
Global Buffer	32 KB	0.061	6.365
Total	-	**20.03**	**223.7**

(aka. multiple-time-step) asynchronously. All the neuron activations are based on the proposed bipolar-based, ST-BIF neuron inference with 2-bit signed input and 8-bit signed weight. For CIFAR-10 datasets, the neuron inference with 4-bit signed weight provides fair comparisons between SOTA. The throughput (TOP/S) is evaluated using a similar method as Neuro-CIM [7]. As recent works [7]–[9] on SNN architecture and circuits focus on IF and LIF SNN activation, it is not fair to compare them with BiNeuroRAM bipolar SNN. Therefore, we design an ADC-based [28] bipolar neuromorphic accelerator (BiNeuroRAM without LPVSA & STC) as the baseline for further ablation studies. Furthermore, we mainly compare the overall performance of BiNeuroRAM with SOTA SNN accelerators (TrueNorth [2], Neuro-CIM [7], NeuRRAM [9], and Tempo-CIM [8]).

C. Performance Comparison

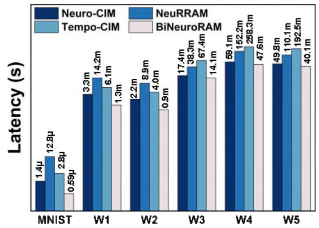
Table II summarizes the performance with SOTA accelerators (TrueNorth, Neuro-CIM, NeuRRAM, and Tempo-CIM). In contrast, BiNeuroRAM is the only design that supports bipolar neuron polarity and benefits from high inference accuracy with low latency. The energy efficiency gain of BiNeuroRAM comes from the low-power design of our LPVSA and STC with high sparsity SNN operation.

1) Inference Accuracy & Latency: As discussed previously, the bipolar neuron model of ST-BIF can mitigate the accuracy drop of SNN converted by the ANN-to-SNN algorithm, in return for a promising inference accuracy. In Fig. 7, BiNeuroRAM outperforms SOTA works in terms of accuracy and latency with different datasets. Moreover, our results for MNIST dataset with 3-layer MLP reaches an accuracy of 99.2%, CIFAR-10 dataset with ResNet-20 can achieve an accuracy of 89.2%, and ImageNet dataset with ResNet-50 achieve an accuracy of 80.9%. Moreover, the latency of SOTAs is evaluated using normalized results based on the throughput ratio. The result in Fig. 7b shows that our latency for different inference workloads is the lowest among SOTAs.

2) Energy Breakdown: Fig. 8 shows the energy breakdown of BiNeuroRAM for different models and datasets. STC's highest energy consumption in every evaluation is primarily due to its extended operation period and elevated power requirements. The significant power demand of STC also results in a high toggling rate for generating pulses. Although with the high power requirement, Fig. 9a addresses STC as the better solution than ADC in our architecture. ReRAM and LPVSA have the second-highest energy consumption,

Dataset	Model	Accelerator	Acc (%)
MNIST	7-layer	NeuRRAM	99.0
	3-layer	Tempo-CIM	96.9
	3-layer	**Our Work**	**99.2**
CIFAR-10	ResNet-12	Neuro-CIM	86.4
	ResNet-12	**Our Work(W1)**	**93.7**
	ResNet-20	NeuRRAM	85.7
	ResNet-20	**Our Work(W2)**	**89.2**
ImageNet	ResNet-18	**Our Work(W3)**	**69.5**
	ResNet-34	**Our Work(W4)**	**72.2**
	ResNet-50	Neuro-CIM	72.5
	ResNet-50	**Our Work(W5)**	**80.9**

(a) Accuracy



(b) Latency

Fig. 7: BiNeuroRAM (a) accuracy and (b) latency compared with SOTA in different models and datasets.

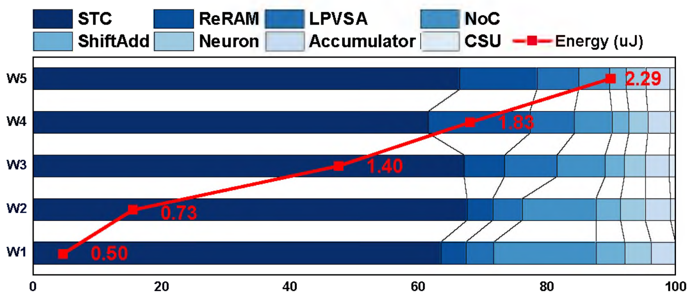


Fig. 8: BiNeuroRAM normalized energy breakdown over different models and datasets.

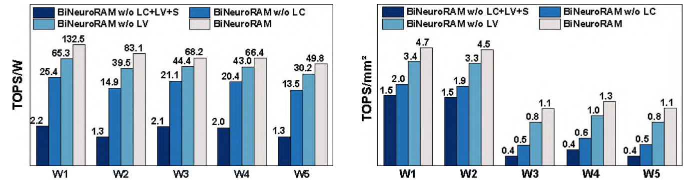
primarily due to their support for constant read operations on large numbers of small-size ReRAM arrays. Furthermore, a large percentage of power from NoC is due to the high static power of the router. The figure also demonstrates the total energy consumption when inference using different datasets and models. BiNeuroRAM achieves energy consumption of 0.5μ - 0.73μJ for CIFAR-10 classification and 1.4μ - 2.29μJ for ImageNet classification.

3) Performance Analysis: The overall performance for energy efficiency and throughput density of BiNeuroRAM design is tabulated in Table II. In addition, Fig. 9 illustrates the energy efficiency and throughput density over different datasets and evaluation models respectively. When performing CIFAR-10 classification with ResNet-12, BiNeuroRAM can achieve 132.5TOPS/W, 1.1× higher than Neuro-CIM. BiNeuroRAM also outperforms Neuro-CIM by 2.1× for ImageNet classification with our energy efficiency up to 68.2TOPS/W at ResNet-18. Our throughput density varies across different datasets for up to 4.72TOPS/mm^2, achieving 2.1×, 16.9×, and 6743× breakthrough with Neuro-CIM, Tempo-CIM, and TrueNorth, respectively.

D. Ablation Study

We conduct an ablation study to analyze the benefits of *STC* & *LPVSA* circuit design for improving energy efficiency. Therefore, we design an ADC to replace STC and LVPSA in BiNeuroRAM as the baseline design. The comparison results are illustrated in Fig. 9. On average, the energy efficiency and throughput density of BiNeuroRAM can be improved by 46.1× and 3.1× respectively over ADC-based design.

Furthermore, we focus on the benefits of LPVSA. Compared to traditional CSA [30] and voltage sense amplifier (VSA) [29], our LPVSA can reduce the ReRAM read power by 58.2× and 14.7× respectively. LPVSA also able to reduce area overhead by 24× and 8.8× respectively. In the system level performance evaluation in Fig. 9, BiNeuroRAM with LPVSA can improve average by 4.2× and 2.3× for CSA, 1.8× and 1.4× for VSA of energy efficiency and throughput



(a) Energy Efficiency (b) Throughput Density

Fig. 9: BiNeuroRAM (a) energy efficiency and (b) throughput density performance compared without LPVSA (replaced with VSA [29] - LV; & replaced with CSA [30] - LC), and STC (replaced with ADC [28] - S).

TABLE II: Comparison with previous SNN chip designs.

	TrueNorth TCAD'15 [2]	Neuro-CIM JSSC'23 [7]	NeuRRAM Nature'22 [9]	Tempo-CIM JETCAS'23 [8]	**BiNeuroRAM This Work**
Application	SNN	SNN	ANN/SNN	SNN	SNN
Technology	28nm CMOS	28nm CMOS	130nm CMOS	40nm CMOS	28nm CMOS
Bitcell	12T SRAM	8T SRAM	ReRAM	ReRAM	ReRAM
Macro Size	16x256	64x160	256x256	256x64	1024x1024
Neuron operation	Digital	Charge-based Mixed-Signal	Analog	Charge Pump -based Analog	Charge-based Mixed-Signal
Area (mm^2)	430	2.9	159	0.084	20.03
Activation Precision	-	1~8	1/2/4/8	4	2
Weight Precision	-	1/4/8	4	2/4/8	1~8
Inference Accuracy (MNIST)	-	-	99.0$^{\triangleleft}$(↓0.2%)	96.9$^{\times}$(↓2.3%)	**99.2$^{\times}$**
Inference Accuracy (CIFAR-10@RN-12)	-	86.4 (↓7.3%)	-	-	**93.7**
Inference Accuracy (CIFAR-10@RN-20)	-	-	85.7 (↓3.5%)	-	**89.2**
Inference Accuracy (ImageNet@RN-50)	-	72.5 (↓8.4%)	-	-	**80.9**
Energy Efficiency (TOPS/W)	0.4	32.61† - 124.15°	67.8$^{\star\circledast}$ - 416.5$^{\star\circledast}$	168.4$^{\star\times}$	**49.8‡ - 132.5$^{\circ\ddagger}$**
Throughput Density (TOPS/mm^2)	0.0007	2.26°	0.0079$^{\star\circledast}$ - 0.0622$^{\star\circledast}$	0.28$^{\star\times}$	**1.15‡ - 4.72$^{\circ\ddagger}$**
Neuron Type	20 Types	IF	IF	LIF	**IF, ST-BIF**

$\star$: Normalized to 28nm and 0.9V based on $P \propto CV^2$, for fair comparison; †: ImageNet @ ResNet-18
$\circ$: CIFAR-10 @ ResNet-12; ×: MNIST 3-Layers MLP; $\triangleleft$: MNIST 7-Layers MLP;
$\circledast$: Measured for performing 256×256 matrix-vector multiplications;
‡: Evaluate using similar method as Neuro-CIM [7] (# of op = 2×# of accumulations/time window)

density, respectively. In short, the proposed LPVSA and STC show a significant design for energy-efficient SNN PIM.

V. Conclusion

We present a high-accuracy bipolar-based neuromorphic design using emerging NVM ReRAM sense via our energy efficient LPVSA, namely BiNeuroRAM, with asynchronous microarchitecture and sparsity optimization techniques to improve energy efficiency. The design of LPVSA significantly improved overall system-level energy performance by enabling the implementation of small array size, which increased compute parallelism. Overall, BiNeuroRAM represents a significant breakthrough in integrating bipolar SNN neurons with PIM architecture, achieving up to 7.3% and 8.4% higher accuracy on the CIFAR-10 and ImageNet datasets, respectively, compared to SOTA SNN PIM designs. Additionally, it delivers energy efficiency improvements of up to 1.1x and 2.1x for CIFAR-10 and ImageNet, respectively, over the current best-performing SNN PIM designs. To our knowledge, BiNeuroRAM is the first bipolar spiking SNN neuromorphic accelerator with asynchronous sparse processing-in-memory, achieving competitive energy efficiency and high accuracy compared to IF neuromorphic hardware designs.

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